Unit 6: Unequal Sized Clusters
Unequal Sized Cluster Sampling

A. Unequal sized clusters
B. Ratio means
C. Illustration

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A. Unequal sized clusters

- Unequal-sized clusters (with subsampling) pose problems for sample selection and estimation
- Fixed two-stage sampling rates f_a=1/2 and f_b=1/10 (i.e., f=1/20) for the following population of A = 10 unequal sized clusters with N=1850:

$$f = f_a \times f_b = \frac{1}{2} \times \frac{1}{10} = \frac{1}{20} \text{ and } f \times N = \frac{1}{20} \times 1850 = 92.5$$

Two unequal cluster samples, *f*=1/20

Cluster	Size	Sample 1			Sample 2		
α	B_α	α	B_α	$x_\alpha = f_b B_\alpha$	α	B_α	$x_\alpha = f_b B_\alpha$
1	400	1	400	40			
2	310				2	310	31
3	40	3	40	4			
4	150				4	150	15
5	250	5	250	25			
6	220				6	220	22
7	50	7	50	5			
8	130				8	130	13
9	90	9	90	9			
10	210				10	210	21
Total	1850	5	830	83	5	1020	102

Variation in sample size

- Sample size is a random variable
  - *y*/*n* is no longer appropriate
  - Ratio mean $r=y/x$ is appropriate
- Alternatives to variation in sample size:
  - Depart from *epsem*
    - Leads to weighting
    - Increase in variance due to weights
  - Create new clusters of equal size
  - Stratify by size
  - Probability proportional to size selection (PPS) [We'll come back to this later]

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B. Ratio means

- Two principle problems with ratio means:
  - Biased for the overall population mean
  - The sampling variance of the ratio mean is not known exactly (except for some special sample designs)
  - Estimate sampling variance using Taylor Series Linearization:

$$r = \frac{y}{x} = \frac{\sum_{\alpha=1}^{a} y_\alpha}{\sum_{\alpha=1}^{a} x_\alpha} = \frac{\sum_{\alpha=1}^{a} \sum_{\beta=1}^{x_\alpha} y_{\alpha\beta}}{\sum_{\alpha=1}^{a} x_\alpha}$$

$$\operatorname{var}(r) \approx \frac{1}{x^2}\left[\operatorname{var}(y) + r^2 \operatorname{var}(x) - 2r\operatorname{cov}(y,x)\right]$$

Variance for sample total

- Recall from equal size one-stage cluster sampling,

$$\mathrm{var}\left(\bar{y}\right)=\frac{1-f}{aB^2\left(a-1\right)}\left(\sum_{\alpha=1}^{a}y_{\alpha}^{2}-\frac{\left(\sum_{\alpha=1}^{a}y_{\alpha}\right)^{2}}{a}\right)$$

- For sample total,

$$\mathrm{var}\left(y\right)=\mathrm{var}\left(aB\times\bar{y}\right)=\left(aB\right)^{2}\mathrm{var}\left(\bar{y}\right)$$

$$=a\frac{1-f}{a-1}\left(\sum_{\alpha=1}^{a}y_{\alpha}^{2}-\frac{y^{2}}{a}\right)$$

Estimating sample total variances

- For an SRS of *a* clusters

$$\mathrm{var}(y) = a\frac{(1-f)}{a-1}\left(\sum_{\alpha=1}^{a} y_\alpha^2 - \frac{\left(\sum_{\alpha=1}^{a} y_\alpha\right)^2}{a}\right)$$

$$\mathrm{var}(x) = a\frac{(1-f)}{a-1}\left(\sum_{\alpha=1}^{a} x_\alpha^2 - \frac{\left(\sum_{\alpha=1}^{a} x_\alpha\right)^2}{a}\right)$$

$$\mathrm{cov}(y,x) = a\frac{(1-f)}{a-1}\left(\sum_{\alpha=1}^{a} y_\alpha x_\alpha - \frac{\left(\sum_{\alpha=1}^{a} y_\alpha\right)\left(\sum_{\alpha=1}^{a} x_\alpha\right)}{a}\right)$$

Estimating var(r)

- Substituting into var(r), and simplifying,

$$\operatorname{var}(r) \approx \frac{1}{x^2}\left[\operatorname{var}(y) + r^2 \operatorname{var}(x) - 2r\operatorname{cov}(y,x)\right] =$$

$$= \left(\frac{1}{x^2}\right) a \frac{(1-f)}{(a-1)}\left[\left(\sum_{\alpha=1}^{a} y_\alpha^2 - \frac{y^2}{a}\right) + r^2\left(\sum_{\alpha=1}^{a} x_\alpha^2 - \frac{x^2}{a}\right) - 2r\left(\sum_{\alpha=1}^{a} y_\alpha x_\alpha - \frac{yx}{a}\right)\right]$$

$$= \left(\frac{1}{x^2}\right) a \frac{(1-f)}{(a-1)}\left[\sum_{\alpha=1}^{a} y_\alpha^2 + r^2 \sum_{\alpha=1}^{a} x_\alpha^2 - 2r \sum_{\alpha=1}^{a} y_\alpha x_\alpha\right]$$

Adequacy of approximation

- Since var(r) is an approximation, it would be useful to an indication of when it might fail
- Examination of the Taylor series shows the adequacy of the approximation depends on the coefficient of variation of the sample size:

$$cv(x) = \frac{se(x)}{x} = \frac{\sqrt{\mathrm{var}(x)}}{x} = \frac{\sqrt{a\frac{(1-f)}{a-1}\left(\sum_{\alpha=1}^{a} x_\alpha^2 - \frac{\left(\sum_{\alpha=1}^{a} x_\alpha\right)^2}{a}\right)}}{\sum_{\alpha=1}^{a} x_\alpha}$$

Rule of thumb for adequacy and bias

- As long as cv(x) <0.15, the approximation is reasonably accurate
  - Values of cv(x) as large as 0.20 may also be acceptable
- The bias of r also depends on the coefficient of variation of the denominator:

$$\left|\frac{Bias(r)}{se(r)}\right| < cv(x)$$

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C. Illustration: Ratio mean

- In a hospital authority of 5 hospitals, estimate the proportion of outpatient visits due to trauma
  - Element: outpatient visits
  - Estimate: proportion of visits due to trauma
- Select outpatient records in 2 stages:
  - First hospital-days (A = 5 x 365 = 1825)
    - Clusters: days across hospitals
    - Sample of a = 10 hospital days
  - Second, select all records on a selected day

$$f = \frac{10}{1825} \cdot 1 = \frac{10}{1825}$$

Hospital-days

Sample hospital-day α	Total visits x_α	Trauma visits y_α
1	58	40
2	47	16
3	37	8
4	69	27
5	40	10
6	27	18
7	34	17
8	30	12
9	26	16
10	32	16

$$\sum_{\alpha=1}^{a} x_\alpha = 400 \quad \sum_{\alpha=1}^{a} y_\alpha = 180 \quad \sum_{\alpha=1}^{a} x_\alpha^2 = 17778 \quad \sum_{\alpha=1}^{a} y_\alpha^2 = 4018 \quad \sum_{\alpha=1}^{a} x_\alpha y_\alpha = 7983$$

Inference

- For r = 0.45, estimate var(r), ignoring the fpc:

$$\operatorname{var}(r) \approx \frac{1}{x^2}\frac{a}{a-1}\left[\sum_{\alpha=1}^{a} y_\alpha^2 + r^2\sum_{\alpha=1}^{a} x_\alpha^2 - 2r\sum_{\alpha=1}^{a} y_\alpha x_\alpha\right] =$$

$$= \frac{1}{400^2}\frac{10}{10-1}\left[4018 + 0.45^2 \times 17788 - 2 \times 0.45 \times 7983\right] = 0.003023$$

- For $t_{(0.975,\,9)}$=2.262, 95% C.I. (0.3256, 0.5744)
- Adequacy of the approximation:

$$cv(x) = \frac{\sqrt{\frac{a}{a-1}\left(\sum_{\alpha=1}^{a} x_\alpha^2 - \frac{x^2}{a}\right)}}{\sum_{\alpha=1}^{a} x_\alpha} = \frac{\sqrt{\frac{10}{10-1}\left(17788 - \frac{400^2}{a}\right)}}{400} = 0.1114$$

Design of new sample

- *deff* and *roh*:

$$\text{var}_{srs}\left(r\right) = \frac{r\left(1-r\right)}{n-1} = \frac{\left(0.45\right)\left(0.55\right)}{399} = 0.0006203$$

$$deff = \frac{\text{var}\left(r\right)}{\text{var}_{SRS}\left(r\right)} = \frac{0.003023}{0.0006203} = 4.874$$

$$roh = \frac{deff - 1}{\bar{b} - 1} = \frac{4.874 - 1}{40 - 1} = 0.09934$$

$$\text{where } \bar{b} = \frac{400}{10} = 40\left(= \frac{x}{a}\right)$$

- What would happen if $a = 10$ and $\bar{b} = 20$?
  - Compute new *deff*, and multiply by SRS sampling variance

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B. Variance estimation

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A. Principles of selection

- Cluster sample stratification similar to elements
  - Use cluster characteristics
  - Homogeneous, mutually exclusive, exhaustive
  - Control the distribution of the sample
  - Decrease sampling variance
  - Selection of stratifying variables, boundaries, etc., discussed for element sampling applies to clusters
- Allocation of sample clusters across strata:
  - Proportionate allocation
  - Paired selection
  - Other allocations

Notation

- Add subscript to denote cluster stratification
- Population of N elements, $B_{h\alpha}$ per cluster
- A_h clusters per stratum
- Population mean: $\bar{Y} = \sum_{h=1}^{H} \sum_{\alpha=1}^{A_h} \sum_{\beta=1}^{B_{h\alpha}} Y_{h\alpha\beta} \Big/ N$

Sample estimation

- At the sample level, $y_{h\alpha\beta}$ denotes survey variable values
- Selection of a_h from A_h clusters (with or without replacement)
- Selection of $b_{h\alpha}$ elements from $(h\alpha)^{th}$ cluster
- Simple replicated subsampling, or ultimate cluster sampling, within each stratum

$$r = \frac{y}{x} = \frac{\sum_{h=1}^{H}\sum_{\alpha=1}^{a_h}\sum_{\beta=1}^{b_{h\alpha}} y_{h\alpha\beta}}{\sum_{h=1}^{H}\sum_{\alpha=1}^{a_h}\sum_{\beta=1}^{b_{h\alpha}} x_{h\alpha\beta}} = \frac{\sum_{h=1}^{H}\sum_{\alpha=1}^{a_h} y_{h\alpha}}{\sum_{h=1}^{H}\sum_{\alpha=1}^{a_h} x_{h\alpha}} = \frac{\sum_{h=1}^{H} y_h}{\sum_{h=1}^{H} x_h}$$

Weighted estimation

- The quantities $y_{h\alpha}$ and $x_{h\alpha}$ can be weighted, if not *epsem*
  - *r* implicitly uses $y^{*}_{h\alpha\beta} = w_{h\alpha\beta} y_{h\alpha\beta}$
  - $w_{h\alpha\beta}$ is the weight for the $(h\alpha\beta)^{th}$ element
  - $y_{h\alpha} = \sum_{\beta=1}^{b_{h\alpha}} w_{h\alpha\beta} y_{h\alpha\beta} = \sum_{\beta=1}^{b_{h\alpha}} y^{*}_{h\alpha\beta}$ is the weighted estimate for the primary selection in stratum *h*
- "Best" estimate of the total $Y_{h\alpha}$
- Similarly, $x^{*}_{h\alpha\beta} = w_{h\alpha\beta} x_{h\alpha\beta}$ and $x_{h\alpha} = \sum_{\beta=1}^{b_{h\alpha}} w_{h\alpha\beta} x_{h\alpha\beta} = \sum_{\beta=1}^{b_{h\alpha}} x^{*}_{h\alpha\beta}$

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B. Variance estimation

- Sampling variance estimation depends on the methods used to select the clusters within each stratum
  - Based on ultimate cluster sampling theory
  - Under disproportionate allocation, (1 - f_h) appears in each stratum
- Three different methods
  - Multiple differences
  - Paired selection
  - Successive differences

Multiple differences

$$\operatorname{var}(r) \approx \frac{1}{x^2}\left[\operatorname{var}(y) + r^2 \operatorname{var}(x) - 2r\operatorname{cov}(y,x)\right]$$

$$= \frac{1}{x^2}\left[\sum_{h=1}^{H}\operatorname{var}(y_h) + r^2\sum_{h=1}^{H}\operatorname{var}(x_h) - 2r\sum_{h=1}^{H}\operatorname{cov}(y_h, x_h)\right]$$

$$= \frac{1}{x^2}\left[\begin{array}{l} \displaystyle\sum_{h=1}^{H}\frac{a_h(1-f_h)}{a_h-1}\left(\sum_{\alpha=1}^{a_h} y_{h\alpha}^2 - \frac{y_h^2}{a_h}\right) \\ \displaystyle +r^2\sum_{h=1}^{H}\frac{a_h(1-f_h)}{a_h-1}\left(\sum_{\alpha=1}^{a_h} x_{h\alpha}^2 - \frac{x_h^2}{a_h}\right) \\ \displaystyle -2r\sum_{h=1}^{H}\frac{a_h(1-f_h)}{a_h-1}\left(\sum_{\alpha=1}^{a_h} y_{h\alpha}x_{h\alpha} - \frac{y_h x_h}{a_h}\right) \end{array}\right]$$

Paired selection

- a_h = 2 for all h

$$\text{var}(r) \approx \frac{1}{x^2}\left[\begin{array}{l}\sum_{h=1}^{H}(1-f_h)(y_{h1}-y_{h2})^2 + r^2\sum_{h=1}^{H}(1-f_h)(x_{h1}-x_{h2})^2 \\ -2r\sum_{h=1}^{H}(1-f_h)(y_{h1}-y_{h2})(x_{h1}-x_{h2})\end{array}\right]$$

- Selection may be
  - Paired selection design
  - Systematic selection collapsed to paired differences
  - One selection per stratum collapsed to pairs

Successive differences

- Systematic selection from ordered list
- Successive differences ($a_h > 2$ for all h):

$$\mathrm{var}(r) \approx \frac{1}{x^2}\left[\begin{array}{l} \displaystyle\sum_{h=1}^{H} \frac{a_h(1-f_h)}{2(a_h-1)} \sum_{g=1}^{a_h-1} \left(y_{hg} - y_{h,g+1}\right)^2 \\ \displaystyle +r^2 \sum_{h=1}^{H} \frac{a_h(1-f_h)}{2(a_h-1)} \sum_{g=1}^{a_h-1} \left(x_{hg} - x_{h,g+1}\right)^2 \\ \displaystyle -2r \sum_{h=1}^{H} \frac{a_h(1-f_h)}{2(a_h-1)} \sum_{g=1}^{a_h-1} \left(y_{hg} - y_{h,g+1}\right)\left(x_{hg} - x_{h,g+1}\right) \end{array}\right]$$

Alternative formulation

- These formulas can also be expressed in terms of z_{ha}
- Reflect an underlying bivariate regression of y on x through the origin
- $z_{h\alpha} = y_{h\alpha} - rx_{h\alpha}$

Hospital example revisited

Sample hospital-day α	Total visits x_α	Trauma visits y_α	$z_\alpha = y_\alpha - r x_\alpha$
1	58	40	13.9
2	47	16	-5.2
3	37	8	-8.7
4	69	27	-4.1
5	40	10	-8.0
6	27	18	5.9
7	34	17	1.7
8	30	12	-1.5
9	26	16	4.3
10	32	16	1.6

Three formulas revisited

$$\operatorname{var}(r) \approx \frac{1}{x^2}\left[\sum_{h=1}^{H} \frac{a_h(1-f_h)}{a_h-1}\left(\sum_{\alpha=1}^{a_h} z_{h\alpha}^2 - \frac{z_h^2}{a_h}\right)\right]$$

$$\operatorname{var}(r) \approx \frac{1}{x^2}\left[\sum_{h=1}^{H} (1-f_h)\left(z_{h1}-z_{h2}\right)^2\right]$$

$$\operatorname{var}(r) \approx \frac{1}{x^2}\left[\sum_{h=1}^{H} \frac{a_h(1-f_h)}{2(a_h-1)} \sum_{g=1}^{a_h-1}\left(z_{hg}-z_{h,g+1}\right)^2\right]$$

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A. Illustration
B. Probabilities of selection
C. Weighted mean
D. Increase in variance

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A. Illustration

- Select adults in public housing apartments
- Study problem of crime victimization among public housing residents
- Select a sample of a = 10 apartments from master list provided by the public housing authority
  - Among A_1 = 300 apartments east of the expressway, select a_1 = 5 apartments and adults
  - Among A_2 = 100 apartments west of the expressway, select a_2 = 5 apartments and adults
- Within apartments, list adults, select 1 at random
- Obtain a report of the number of times selected adult was victimized in the last 12 months

Sample data

Stratum h	Apt. α	No. adults $B_{h\alpha}$	Resp. $y_{h\alpha\beta}$	Weight $w_{h\alpha\beta}$	Wgtd. Resp. $w_{h\alpha\beta}y_{h\alpha\beta}$
East	1	2	2	120	240
	2	1	0	60	0
	3	3	1	180	180
	4	2	4	120	480
	5	1	2	60	120
West	1	1	0	20	0
	2	2	1	40	40
	3	2	1	40	40
	4	3	5	60	300
	5	2	2	40	80

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B. Probabilities of selection

- Probabilities of selection vary by strata and apartments:

$$\Pr\{h\alpha\beta\} = \Pr\{h\alpha\} \times \Pr\{\beta \mid h\alpha\}$$

  - h denotes the "side" of the expressway
  - α denotes the apartment
  - β denotes the adult within the apartment

- Alternatively,

$$\Pr\{\text{selecting adult}\} = \Pr\{\text{selecting apartment}\} \times \Pr\{\text{selecting adult} \mid \text{apartment selected}\}$$

- Compensatory weight the inverse: $w_{h\alpha\beta} = 1 / \Pr\{h\alpha\beta\}$

Sample data

Stratum h	Apt. α	No. adults $B_{h\alpha}$	Resp. $y_{h\alpha\beta}$	Weight $w_{h\alpha\beta}$	Wgtd. Resp. $w_{h\alpha\beta}y_{h\alpha\beta}$
East	1	2	2	120	240
	2	1	0	60	0
	3	3	1	180	180
	4	2	4	120	480
	5	1	2	60	120
West	1	1	0	20	0
	2	2	1	40	40
	3	2	1	40	40
	4	3	5	60	300
	5	2	2	40	80

Two apartments

- For an east-side apartment, $\Pr\{11\beta\} = \Pr\{11\} \times \Pr\{\beta \mid 11\} = \dfrac{5}{300} \times \dfrac{1}{B_{11}}$
  - B_{11} = number of adults in apartment {11} = 2
  - $\Pr\{11\beta\} = \dfrac{5}{300} \times \dfrac{1}{2} = \dfrac{1}{120}$
  - Weight

    $w_{11\beta} = (1/120)^{-1} = 120$

- Similarly, $\Pr\{21\beta\} = \Pr\{21\} \times \Pr\{\beta \mid 21\} = \dfrac{5}{100} \times \dfrac{1}{B_{21}}$
- Weight $w_{21\beta} = (1/20)^{-1} = 20$

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C. Weighted mean

- The weighted estimator of the mean is

$$\bar{y}_w = \frac{\sum_{h=1}^{H}\sum_{\alpha=1}^{a_h}\sum_{\beta=1}^{b_{h\alpha}} w_{h\alpha\beta}\, y_{h\alpha\beta}}{\sum_{h=1}^{H}\sum_{\alpha=1}^{a_h}\sum_{\beta=1}^{b_{h\alpha}} w_{h\alpha\beta}} = \frac{1480}{740} = 2.0$$

- In contrast, the unweighted estimate is

$$\bar{y} = \frac{\sum_{h=1}^{H}\sum_{\alpha=1}^{a_h}\sum_{\beta=1}^{b_{h\alpha}} y_{h\alpha\beta}}{n} = \frac{18}{10} = 1.8$$

Variance estimation

- Variance estimation takes into account the fact that the denominator varies
- Depends on which sample of apartments selected
- Weighted mean is a ratio of random variables:

$$\bar{y}_w = \frac{\sum_{h=1}^{H}\sum_{\alpha=1}^{a_h}\sum_{\beta=1}^{b_{h\alpha}} w_{h\alpha\beta}\, y_{h\alpha\beta}}{\sum_{h=1}^{H}\sum_{\alpha=1}^{a_h}\sum_{\beta=1}^{b_{h\alpha}} w_{h\alpha\beta}} = \frac{\sum_{h=1}^{H}\sum_{\alpha=1}^{a_h} u_{h\alpha}}{\sum_{h=1}^{H}\sum_{\alpha=1}^{a_h} w_{h\alpha}} = \frac{u}{w}$$

Taylor series approximation

- The approximation is

$$\operatorname{var}(\bar{y}_w) \approx \frac{1}{w^2}\left[\operatorname{var}(u) + r^2 \operatorname{var}(w) - 2r\operatorname{cov}(u,w)\right]$$

- For stratified clusters, ignoring the fpc,

$$\operatorname{var}(u) = \sum_{h=1}^{H}\left(\frac{a_h}{a_h - 1}\right)\left(\sum_{\alpha=1}^{a_h} u_{h\alpha}^2 - \frac{u_h^2}{a_h}\right)$$

$$\operatorname{var}(w) = \sum_{h=1}^{H}\left(\frac{a_h}{a_h - 1}\right)\left(\sum_{\alpha=1}^{a_h} w_{h\alpha}^2 - \frac{w_h^2}{a_h}\right)$$

$$\operatorname{cov}(w,u) = \sum_{h=1}^{H}\left(\frac{a_h}{a_h - 1}\right)\left(\sum_{\alpha=1}^{a_h} w_{h\alpha}u_{h\alpha} - \frac{w_h u_h}{a_h}\right)$$

Sample data

Stratum	Apt. α	Wght. $w_{h\alpha}$	$u_{h\alpha}$	$w_{h\alpha}^2$	$u_{h\alpha}^2$	$w_{h\alpha}u_{h\alpha}$
East	1	120	240	14400	57600	28800
	2	60	0	3600	0	0
	3	180	180	32400	32400	32400
	4	120	480	14400	230400	57600
	5	60	120	3600	14400	7200
Subtotal		540	1020	68400	334800	126000
West	1	20	0	400	0	0
	2	40	40	1600	1600	1600
	3	40	40	1600	1600	1600
	4	60	300	3600	90000	18000
	5	40	80	1600	6400	3200
Subtotal		200	460	8800	99600	24400
Total	10	740	1480	77200	434400	150400

Variance of sample totals

$$\operatorname{var}(u)=\sum_{h=1}^{H}\left(\frac{a_h}{a_h-1}\right)\left(\sum_{\alpha=1}^{a_h}u_{h\alpha}^2-\frac{u_h^2}{a_h}\right)$$

$$=\left(\frac{5}{4}\right)\left(334800-\frac{1020^2}{5}\right)+\left(\frac{5}{4}\right)\left(99600-\frac{460^2}{5}\right)=230,000$$

$$\operatorname{var}(w)=\sum_{h=1}^{H}\left(\frac{a_h}{a_h-1}\right)\left(\sum_{\alpha=1}^{a_h}w_{h\alpha}^2-\frac{w_h^2}{a_h}\right)$$

$$=\left(\frac{5}{4}\right)\left(68400-\frac{540^2}{5}\right)+\left(\frac{5}{4}\right)\left(8800-\frac{200^2}{5}\right)=13,600$$

$$\operatorname{cov}(u,w)=\sum_{h=1}^{H}\left(\frac{a_h}{a_h-1}\right)\left(\sum_{\alpha=1}^{a_h}u_{h\alpha}w_{h\alpha}-\frac{u_h w_h}{a_h}\right)$$

$$=\left(\frac{5}{4}\right)\left(126000-\frac{540\times1020}{5}\right)+\left(\frac{5}{4}\right)\left(24400-\frac{200\times460}{5}\right)=27,300$$

Ratio mean estimated variance

- Combining these estimates,

$$\operatorname{var}\left(\bar{y}_w\right) \approx \frac{1}{740^2}\left[230,000+\left(2.0\right)^2\left(13,600\right)-2\left(2.0\right)\left(27,300\right)\right]$$

$$= 0.3199$$

$$se\left(\bar{y}_w\right) = 0.5656$$

- For confidence intervals, *df* = *a* - *H*

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D. Increase in variance

- Weighting can increase sampling variance of estimates
- Assuming “arbitrary” weighting
- Potential increase in variance due to weighting:

$$1+L=\frac{n\left(\sum_{i=1}^{n} w_i^2\right)}{\left(\sum_{i=1}^{n} w_i\right)^2}=\frac{10\left(77200\right)}{\left(740\right)^2}=1.410$$

- Increase depends on variability of weights.

Unit 6: Unequal Sized Clusters PPS

A. Controlling sample size
B. PPS
C. Systematic PPS

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A. Controlling sample size

- Potentially large variation in sample size
  - A ratio mean for design-consistent estimation
- More control over sample size. Why?
  - Bias of r
  - Variance of r
- Equalize workloads controlling survey costs

Example: Two-stage selection

- A sample of n = 36 employees from N = 2,160
  - f = 36/2160 = 1/60

Unit	No.	Unit	No.
1	443	6	291
2	162	7	64
3	127	8	70
4	554	9	232
5	115	10	102

Sample size control

- Redefine clusters
- Stratification by size, varying sampling rates by size of cluster stratum
- Give up *epsem* for equal size subsamples
- Probabilities proportionate to size (PPS)

Redefine clusters

- Create 10 clusters of approximately $N/10 = 216$ employees each
- Variation above and below desired cluster size
- Once clusters created, select the sample with uniform sampling fractions

$$f = f_a f_b = \left(\frac{a}{A}\right)\left(\frac{b}{B}\right)$$

$$fN = \left(\frac{1}{60}\right)(2160) = 36$$

Stratify by size

Stratum I		Stratum II	
Unit	No.	Unit	No.
2	162	1	443
3	127	4	554
5	115	6	291
7	64	9	232
8	70		1520
10	102		
	640		

Varying sampling rates

- Obtain *epsem* by choice of first and second stage rates:
- Use in stratum I: $f_{Ia} = \frac{1}{6} \qquad f_{Ib} = \frac{1}{10}$
- Use in stratum II: $f_{IIa} = \frac{1}{4} \qquad f_{IIb} = \frac{1}{15}$
- Sample size in stratum I ranges from 6 to 17, stratum II from 15 to 37
- cv(x) has been reduced within strata.

Equal subsample size

- Select clusters with equal rates but take equal size subsamples
- Non-*epsem* design

$$f = f_a f_{b|a} = \left(\frac{a}{A}\right)\left(\frac{b}{B_\alpha}\right) = \left(\frac{2}{10}\right)\left(\frac{18}{B_\alpha}\right) = \frac{16}{10 \times B_\alpha} = f_\alpha$$

- Knowing B_α for each selected cluster, create weights $w_{\alpha\beta} = f_\alpha^{-1}$
- Constant overall workloads at the price of weighting and increases in sampling variances relative to *epsem* designs

Unit 6: Unequal Sized Clusters PPS

A. Controlling sample size
B. PPS
C. Systematic PPS

Unit 6: Unequal Sized Clusters
PPS

A. Controlling sample size
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PPS, or how to achieve *epsem* and equal subsample sizes at the same time

- Goal: Select a two stage sample such that there are equal subsample sizes in each cluster with *epsem*
- We want an overall *epsem* rate $f = n/N$ such that the sampling rate in the second stage is b/B_α

$$f = f_\alpha f_{\beta|\alpha} = \left(f_\alpha\right)\left(\frac{b}{B_\alpha}\right) = \left(\frac{n}{N}\right)$$

- Solving $f_\alpha = \frac{n}{N} \times \frac{B_\alpha}{b} = \frac{ab}{\sum_{\alpha=1}^{A} B_\alpha} \times \frac{B_\alpha}{b} = \frac{aB_\alpha}{\sum_{\alpha=1}^{A} B_\alpha}$

PPS implications

- Select clusters with probabilities proportionate to their size, B_α
  - Overall the selection becomes $f = \dfrac{aB_\alpha}{\sum_{\alpha=1}^{A} B_\alpha} \times \dfrac{b}{B_\alpha}$
- Selection probability varies at both first & second stages
  - Overall, *epsem*
  - Large clusters selected with high probabilities, *subsampled* at low rate
  - Small clusters selected with low probabilities, *subsampled* at high rate

Selecting clusters with PPS

- Two approaches
  - Simple replicated subsampling – at random and with replacement
  - Systematic sample and without replacement
- For both approaches, need to cumulate (sequentially add up) the size of the clusters

Selecting clusters with PPS

Unit	B_α	Cum. B_α	Unit	B_α	Cum. B_α
1	443	443	6	291	1692
2	162	605	7	64	1756
3	127	732	8	70	1826
4	554	1286	9	232	2058
5	115	1401	10	102	2160

Using cumulative counts

- Simple replicated subsampling for a=2:
  - Select RN from 1 to 2160 with replacement
  - Find the cluster (Unit) with cumulative sum first exceeding RN
  - Select
  - Repeat
  - Subsampling rate b / B_α requires exactly b
- If cluster selected k times, take $k \times b$ elements
- Subsampling: SRS, systematic, stratified element, even cluster sampling

Calculating subsampling rates

- For example, RN_1=702 and RN_2 = 1744 select Units 3 and 7

- For Unit 3, $\dfrac{aB_\alpha}{N} = \dfrac{2 \times 127}{2160} = \dfrac{1}{8.504}$

- Yields within $\dfrac{b}{B_\alpha} = \dfrac{18}{127} = \dfrac{1}{7.056}$

Variance estimation

- For PPS, $n=ab$ is fixed. Hence,

$$\bar{y} = \frac{\sum_{\alpha=1}^{a} \sum_{\beta=1}^{b} y_{\alpha\beta}}{ab}$$

$$\operatorname{var}(\bar{y}) = \frac{(1-f)}{a} s_a^2$$

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C. Systematic PPS

- Compute zone size: N/a = 2160/2 = 1080
  - Select one subsample from each zone size 1080
  - Zone boundary falls 348 "elements" (measure units) through Unit 4; falls in 2 zones
- Select Units by selecting *RS* from 1 to 1080
  - Identify selection from cumulative counts
  - Add the interval k = 1080 to the *RS*
  - Identify next selected cluster
  - Select one subsample from each at the rate b/B_α

Selecting clusters with PPS

Unit	B_α	Cum. B_α	Unit	B_α	Cum. B_α
1	443	443	6	291	1692
2	162	605	7	64	1756
3	127	732	8	70	1826
4	554	1286	9	232	2058
5	115	1401	10	102	2160

Within sampling rates

- For example, suppose RN = 804
  - Select α = 4
  - $RN + I$ = 804 + 1080 = 1884 selects α = 9

- For α = 4, $\dfrac{aB_\alpha}{N} = \dfrac{B_\alpha}{N/a} = \dfrac{554}{2160/2} = \dfrac{554}{1080} = \dfrac{1}{1.950}$

- Subsample $\dfrac{b}{B_4} = \dfrac{18}{554} = \dfrac{1}{30.78}$

PPS without replacement

- Large literature on without replacement PPS selection procedures
- Alternatives lead to variance estimation expressions that require knowledge about the joint probabilities of selection
- Horvitz-Thompson estimator and the Yates-Grundy estimator of variance
- Unfortunately, the Yates-Grundy estimator can be negative
- Literature proposes wide variety of designs to overcome this problem

Unit 6: Unequal Sized Clusters PPeS

A. Estimated size measures
B. Stratification
C. Oversize units
D. Undersize units

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A. Estimated size measures

- Suppose that the exact count of elements -- B_α -- in each cluster is unknown
- But measure that approximates the number of units (that is, the size) in each cluster -- Measure of Size -- Mos_α
- Example: Do not know the current exact count of employees for each Unit
  - Do know the number counted for each Unit at the last payroll one month ago

Example

Unit	Last payroll	Now
1	443	460
2	162	172
3	127	130
4	554	554
5	115	125
6	291	310
7	64	68
8	70	74
9	232	246
10	102	141
Total	2160	2280

PPeS

- Probabilities proportionate to estimated size: *epsem* design in two stages
- Over-sample large clusters relative to small
- Select clusters with the rate $\dfrac{aMos_{\alpha}}{\sum_{\alpha=1}^{A} Mos_{\alpha}}$
- Then subsample elements at rate $\dfrac{b^{*}}{Mos_{\alpha}}$
- Why $b*$?
  - The actual number of second stage units selected is unknown; it's a target subsample size
  - If $Mos_{\alpha} = B_{\alpha}$, then $b*=b$

Selecting units with PPeS

- Returning to employees example, suppose $b^*=18$, $a=2$, and
- a = 2 clusters are selected $\sum_{\alpha=1}^{10} Mos_{\alpha} = 2160$
- One subsample of expected size b^* = 18 each

$$f = \frac{2Mos_{\alpha}}{\sum Mos_{\alpha}} \times \frac{18}{Mos_{\alpha}} = \frac{36}{2160} = \frac{1}{60}$$

- With PPeS sampling, subsampling at a specified rate and <u>not</u> selecting a <u>fixed</u> number of elements from each selected cluster

Example

Unit	Last payroll	Now
1	443	460
2	162	172
3	127	130
4	554	554
5	115	125
6	291	310
7	64	68
8	70	74
9	232	246
10	102	141
Total	2160	2280

PPeS illustration

- Suppose $RN_1 = 702 \quad RN_2 = 1744$
- Select Units 3 and 7
- For Unit 3,

$$\frac{b}{Mos_3} = \frac{18}{127} = \frac{1}{7.056}$$

- For Unit 7,

$$\frac{b}{Mos_7} = \frac{18}{64} = \frac{1}{3.556}$$

- If $Mos_\alpha = B_\alpha$, a sample of exactly size 36 will be selected

Sample size variation returns

- But here $Mos_{\alpha} \cong B_{\alpha}$
- For Unit 3, $B_{\alpha} = 130$, and $\dfrac{b}{Mos_3} = \dfrac{1}{7.056}$ yields

$$x_a = \frac{b}{Mos_{\alpha}} \times B_{\alpha} = \frac{1}{7.056} \times 130 = 18.425$$

- For interval 7.056, select a sample of 18 employees with probability 0.575
- Or a sample of 19 with probability 0.425

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B. Stratification

- *Mos*, *a*, *b**, etc., are specified within strata

$$f = f_h = \frac{a_h Mos_{h\alpha}}{\sum_{\alpha=1}^{A_h} Mos_{h\alpha}} \times \frac{b_h^*}{Mos_{h\alpha}} = \frac{a_h b_h^*}{\sum_{\alpha=1}^{A_h} Mos_{h\alpha}}$$

- Retain *epsem* for stratified PPeS sampling across strata: $f = f_h$ for all h.

Implicit stratification

- Systematic PPeS sampling implicitly stratifies by selecting within each zone one subsample size b^*
- Stratification notation not necessary with this design
- Zone size (or sampling interval) is

$$\frac{\sum_{\alpha=1}^{A_h} Mos_{h\alpha}}{a_h}$$

Select *a* = 4 units

- a = 4 subsamples of size b^* = 18 from Mos_t = 2160
- Then

$$f=\frac{ab^*}{\sum Mos_\alpha}=\frac{4\times 18}{2160}=\frac{1}{30}$$

- Zone size is 2160/4 = 540

Explicit stratification

Stratum I		Stratum II	
Unit	Mos	Unit	Mos
1	443	5	115
2	162	6	291
3	127	7	64
4	554	8	70
		9	232
		10	102
Total	1286		874

Paired selection

$$f_1 = \frac{1}{30} \neq \frac{2Mos_{1\alpha}}{\sum_{\alpha=1}^{A_1} Mos_{1\alpha}} \frac{18}{Mos_{1\alpha}} = \frac{36}{1286}$$

$$f_2 = \frac{1}{30} \neq \frac{2Mos_{2\alpha}}{\sum_{\alpha=1}^{A_2} Mos_{2\alpha}} \frac{18}{Mos_{2\alpha}} = \frac{36}{874}$$

- Can manipulate $a_h, b_h^*, \sum_{\alpha=1}^{A_h} Mos_{h\alpha}$
- Uncommon to vary a_h

Vary subsample size

- For paired selections,

$$f_h = \frac{2Mos_{h\alpha}}{\sum_{\alpha=1}^{A_h} Mos_{h\alpha}} \frac{b_h^*}{Mos_{h\alpha}}$$

- Adjusting the subsample sizes across strata,

$$f_1 = \frac{2Mos_{1\alpha}}{\sum_{\alpha=1}^{A_1} Mos_{1\alpha}} \frac{b_1^*}{Mos_{1\alpha}} = \frac{1}{30}$$

$$b_1^* = 21.43$$

$$b_2^* = 14.57$$

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Unit 6: Unequal Sized Clusters PPeS

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D. Undersize units

C. Oversize units

- Oversize units are clusters for which $Mos_{h\alpha}$ is so large that the unit has a certain chance of selection
  - Often so large that they will be selected multiple times in systematic PPeS
- For example, Unit 4 has $Mos_4 = 554 > 540$
  - Selected twice with probability

$$\frac{Mos_4}{540} = 1.0259$$

Solutions

- If there are few such clusters and chances of multiple selections small leave them in the list
  - If one selected *k* times, select *k* subsamples
- If there are many such clusters comprising a large share of the population, place in separate strata
  - Each such cluster is now a stratum, a cluster selected with certainty
  - Sampling rate(s) are applied directly within clusters
  - Self-representing units (SRU)

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Unit 6: Unequal Sized Clusters
PPeS

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D. Undersize units

- Undersize units are clusters for which $Mos_{h\alpha} < cb^{*}$
- $Mos_{h\alpha} < b^{*}$ implies sampling within cluster at rate > 1
- All elements must be selected in the cluster, and

$$f = \frac{a_h Mos_{h\alpha}}{\sum Mos_{h\alpha}} \times 1$$

Solutions

- Link undersize units to form linked units of minimum sufficient size
  - Link before selection for the entire frame, especially if the frame is a computerized list
  - Can create clusters with greater heterogeneity
  - In area sampling, link geographically contiguous units
- Linking after selection
- If numerous, place in separate stratum
- Undersize units can include zero measure clusters

Linking after (during) selection

(1) Identify selected unit (block). If the selected unit and next on the list are minimum sufficient size, STOP

(2) If selected unit or next are not of minimum sufficient size,

a) Move forward in list until first unit of minimum sufficient size is encountered

b) Cumulate units backwards until a linked unit of minimum sufficient size is created

c) Continue process until the selected unit is linked

Example

Tract	Block	HU's
28	210	23
	211	17
	212	36
	213	1
29	101	0
	102	0
	103	0
	104	0
	105	1
	106	0
	107	0
	108	0
	109	0
	110	0
	111	0
	112	0
	113	0
	114	60
	115	9

Example

Tract	Block	HU's
28	210	23
	211	17
	212	36
	213	1
29	101	0
	102	0
	103	0
	104	0
	105	1
	106	0
	107	0
	108	0
	109	0
	110	0
	111	0
	112	0
	113	0
	114	60
	115	9

e = 2

Link 101-113, 212-213, size = 38

Link 105-113, size = 1